

Physical-Covariance Block Residual Envelopes

Directional Sharpness versus Global Positive Size

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Abstract

RH-66 repaired catastrophic packetwise scalarization by constructing a block Krylov center and a positive-semidefinite residual Gram envelope. Its trace-global choice of scalar weights was still conservative in a specially cancelling physical coefficient direction. This paper optimizes those weights for a prescribed positive coefficient covariance W .

If $P_1, \dots, P_m \succeq 0$ are the propagated block residual pieces and

$$\mathcal{C}(\theta) = \sum_i \frac{P_i}{\theta_i}, \quad \theta_i > 0, \quad \sum_i \theta_i = 1,$$

then

$$\theta_i^* = \frac{\sqrt{\text{tr}(WP_i)}}{\sum_j \sqrt{\text{tr}(WP_j)}}, \quad \min_{\theta} \text{tr}(W\mathcal{C}(\theta)) = \left(\sum_i \sqrt{\text{tr}(WP_i)} \right)^2.$$

The Young parameter joining center and residual has an analogous exact formula. In the rank-one limit $W = uu^*$, the covariance objective is exactly the RH-66 directional center-radius upper. A strictly positive regularization $W_{\varepsilon} = uu^* + \varepsilon(I - uu^*)$ keeps a globally valid PSD matrix.

We prove a sharpness–size duality. If the physical ray annihilates every residual piece, its excess is $O(\sqrt{\varepsilon})$ while the off-ray global envelope is $\Omega(\varepsilon^{-1/2})$. On the exact cancellation witness, the isotropic physical/global gains are 2.92454/1.16077; at $\varepsilon = 10^{-24}$ they become 1.001027/489.96. Both rows are certified with 256-bit Arb arithmetic. On generic nonnormal and complex-phase models, covariance focusing converges to the directional optima 3.89914 and 1.28847 without a comparable global blow-up.

Factor-first evaluation in the physical coefficient frame removes a binary64 cancellation artifact in RH-66: its archived gain 410.16 is not the mathematical isotropic gain of the exact witness. The block theorem remains valid; the stable factor-first evaluation gives 2.92454. The next gate is uniform block depth and a production derivation of the physical coefficient covariance. No Stage A1 closure, arithmetic trace formula, or Hilbert–Polya conclusion is claimed.

Keywords: coefficient covariance; positive-semidefinite envelope; block Krylov method; phase cancellation; Gram factorization.

MSC 2020: 47A10; 47B65; 65F35; 90C25; 93B07.

1 Introduction

RH-66 produced two block certificates: a sharp center-radius bound for one physical coefficient vector and a PSD Gram bound valid for all packet coefficients [2]. The latter is needed whenever a downstream argument requires positivity before the physical coefficient has been inserted. Its scalar Cauchy and Young weights were chosen to minimize ordinary trace, which treats every coefficient direction equally.

The physical packet fusion is not isotropic. It comes with one selected phase vector, or more generally a small ensemble of admissible phase vectors. This suggests replacing trace by

a positive covariance-weighted trace. The resulting problem is elementary convex optimization [1], but it reveals a useful structural boundary: directional sharpness is compatible with global PSD validity, yet exact annihilation forces the globally valid matrix to become large away from the physical ray.

Contributions and boundary

1. We solve exactly the covariance-weighted residual and Young parameter optimizations.
2. We prove convergence to the rank-one directional certificate and a sharpness–global-size tradeoff for exact residual annihilation.
3. We formulate a factor-first coefficient frame that avoids catastrophic cancellation when Grams contain large opposing entries.
4. We audit one exact and two generic block models, including a 256-bit interval certificate for the exact tradeoff.

The covariance is an input to this paper. It has not yet been derived from the production folded-Gaussian packet family.

2 Covariance-optimal residual weights

RH-66 writes the block remainder Gram as a positive upper built from finitely many pieces $P_i \succeq 0$. The index i includes the propagated Galerkin residuals and the source reconstruction residual. For

$$\theta_i > 0, \text{quad} \sum_{i=1}^m \theta_i = 1, \quad (1)$$

put

$$\mathcal{C}(\theta) = \sum_{i=1}^m \frac{P_i}{\theta_i}. \quad (2)$$

Every such choice is a valid PSD residual envelope.

Theorem 2.1 (Covariance-optimal residual allocation). *Let $W \succ 0$ and discard only identically zero P_i . Define $p_i = \text{tr}(WP_i) > 0$. Then the unique minimizer of $\text{tr}(W\mathcal{C}(\theta))$ under (1) is*

$$\theta_i^* = \frac{\sqrt{p_i}}{\sum_j \sqrt{p_j}}, \quad (3)$$

and

$$\min_{\theta} \text{tr}(W\mathcal{C}(\theta)) = \left(\sum_i \sqrt{p_i} \right)^2. \quad (4)$$

Proof. Weighted Cauchy gives

$$\left(\sum_i \sqrt{p_i} \right)^2 = \left(\sum_i \sqrt{p_i/\theta_i} \sqrt{\theta_i} \right)^2 \leq \sum_i \frac{p_i}{\theta_i}.$$

Equality holds exactly when $\theta_i \propto \sqrt{p_i}$. □

Let $Y \succeq 0$ be the center Gram. RH-66 then uses

$$\mathcal{G}(\eta, \theta) = (1 + \eta)Y + (1 + \eta^{-1})\mathcal{C}(\theta), \quad \eta > 0. \quad (5)$$

Proposition 2.2 (Covariance-optimal Young parameter). *For fixed θ , put $y = \text{tr}(WY)$ and $c = \text{tr}(WC(\theta))$. If $y, c > 0$, then*

$$\eta^* = \sqrt{c/y}, \quad \min_{\eta} \text{tr}(W\mathcal{G}(\eta, \theta)) = (\sqrt{y} + \sqrt{c})^2. \quad (6)$$

The zero cases are obtained as one-sided limits.

Proof. The variable part is $\eta y + c/\eta$, whose arithmetic–geometric mean minimum is $2\sqrt{yc}$. $\square$

These optimizations do not weaken global validity. They select scalar weights inside the same Loewner upper proved by RH-66.

3 Rank-one limit and the global-size price

Let u be a unit physical coefficient vector and $P_{\perp} = I - uu^*$. Consider

$$W_{\varepsilon} = uu^* + \varepsilon P_{\perp}, \quad 0 < \varepsilon \leq 1. \quad (7)$$

Proposition 3.1 (Directional rank-one limit). *If every nonzero residual piece has $u^*P_i u > 0$, then as $\varepsilon \downarrow 0$, the weights in (3), the Young parameter, and the physical value $u^*\mathcal{G}u$ converge to the RH-66 directional center-radius optimum. The envelope remains bounded.*

Proof. All scores $\text{tr}(W_{\varepsilon}P_i) = u^*P_i u + \varepsilon \text{tr}(P_{\perp}P_i)$ converge to strictly positive limits. Hence the formulas in Theorem 2.1 and Proposition 2.2 are continuous. At $W_0 = uu^*$, (4) is precisely the square of the sum of the directional residual radii. $\square$

Exact cancellation is a boundary case, not covered by strict positivity.

Theorem 3.2 (Sharpness–global-size duality). *Assume*

$$P_i u = 0 \quad (1 \leq i \leq m), \quad Yu = Gu, \quad y_0 = u^*Yu > 0, \quad (8)$$

where G is the exact Gram. Suppose the complement residual envelope is nonzero. Then the covariance-optimal residual weights are independent of ε , and

$$\eta_{\varepsilon} = \Theta(\sqrt{\varepsilon}), \quad \frac{u^*\mathcal{G}_{\varepsilon}u}{u^*Gu} = 1 + \Theta(\sqrt{\varepsilon}), \quad \|\mathcal{G}_{\varepsilon}\| = \Omega(\varepsilon^{-1/2}). \quad (9)$$

Proof. Positivity and $u^*P_i u = 0$ imply $P_i u = 0$. Every covariance score is $\varepsilon \text{tr}(P_{\perp}P_i)$, so the common factor cancels from (3); call the resulting nonzero envelope $\mathcal{C}_{\perp}$. Therefore

$$\eta_{\varepsilon} = \sqrt{\frac{\varepsilon \text{tr}(P_{\perp}\mathcal{C}_{\perp})}{y_0 + \varepsilon \text{tr}(P_{\perp}Y)}}.$$

Since $\mathcal{C}_{\perp}u = 0$, the physical value is $(1 + \eta_{\varepsilon})y_0$. On the complement, the term $(1 + \eta_{\varepsilon}^{-1})\mathcal{C}_{\perp}$ has norm of order η_{ε}^{-1} . $\square$

Thus no contradiction exists between a sharp physical ray and a large global PSD matrix. The large off-ray value is the price of requiring one matrix to remain valid for coefficient directions the physical covariance assigns almost zero weight.

4 Factor-first coefficient frames

Let $U = [u, U_\perp]$ be unitary. Mathematically one may transform a Gram by U^*GU . Numerically, this can subtract large nearly equal entries after they have already been squared. The implementation instead transforms each factor first:

$$(XU)^*M(XU), \quad (10)$$

for the exact response, center, and every residual factor. This preserves structural zeros before Gram formation and is substantially more stable in the cancellation witness.

This point corrects a finite-precision diagnostic in RH-66. The archived 410.16 uniform-Gram gain was a valid but severely rounded binary64 upper formed in the original packet coordinates. The exact scalar reduction and 256-bit audit below give the mathematical isotropic gain 2.92454. No analytic RH-66 statement changes.

5 Model audit

The exact witness uses the RH-66 diagonal model. In the physical frame, its target and complement decouple. If y is the exact target energy, $Y_\perp$ the complement center, and $C_\perp$ the optimized residual upper, then

$$\eta_\varepsilon = \sqrt{\frac{\varepsilon C_\perp}{y + \varepsilon Y_\perp}}, \quad \text{physical gain} = 1 + \eta_\varepsilon. \quad (11)$$

Here

$$y = 6.92944 \times 10^{-17}, \quad Y_\perp = 19.7435, \quad C_\perp = 73.1273,$$

so the asymptotic regime begins only below approximately 10^{-18} .

Table 1: Covariance sharpness and global positive size.

model / ε	physical gain	weighted gain	global gain
exact cancellation / 1	2.924544	1.160769	1.160769
exact cancellation / 10^{-24}	1.001027	1.002054	489.960
exact cancellation / 10^{-28}	1.000010	1.000021	4.8933×10^4
nonnormal chain / 1	3.950049	4.004647	3.851919
nonnormal chain / 10^{-10}	3.899145	3.899145	3.934330
complex phase / 1	1.295052	1.343557	1.169230
complex phase / 10^{-10}	1.288471	1.288471	1.154899

The generic models satisfy the strict case of Proposition 3.1. Their physical values converge rapidly to the directional optima 3.8991448 and 1.2884708, while the global spectral gains remain bounded. The computed PSD slack is positive on all rows.

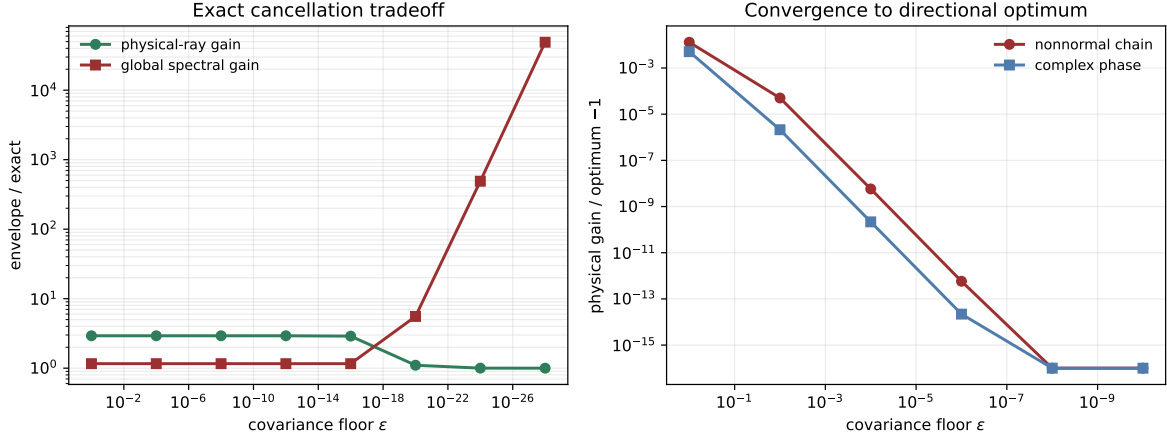


Figure 1: Left: the exact cancellation ray becomes sharp only by enlarging the off-ray global envelope. Right: generic models converge rapidly to their directional optima without that singular tradeoff.

A 256-bit Arb audit certifies the exact scalar ledger. At $\varepsilon = 1$, the physical gain lies in $(2, 3)$ and the global gain is below 1.2. At $\varepsilon = 10^{-24}$, the physical gain lies in $(1.001, 1.002)$ while the global gain exceeds 400.

6 Route consequence

The PSD-envelope wall is now classified rather than merely observed:

1. for generic physical directions, covariance focusing recovers the directional block certificate with bounded global cost;
2. exact annihilation directions force a tradeoff between physical sharpness and global PSD size;
3. stable factor-first coordinates are mandatory before deciding which regime is present.

This leaves two analytic inputs. First, the physical packet construction must supply an admissible covariance W_σ or prove that one selected ray is sufficient downstream. Second, the required block Krylov depth must be controlled uniformly. RH-68 should therefore audit uniform depth and prove a no-go theorem if a growing family of separately visible modes forces the rank to grow.

7 No arithmetic or Hilbert–Polya conclusion

This paper constructs no self-adjoint operator, no $T \log T$ counting law, no prime-power trace formula, and no completed-zeta identity. It makes no Hilbert–Polya or Riemann-hypothesis claim. Stage A1 and unconditional Stage A4 remain open.

8 Reproducibility

The directory contains covariance optimization code, tests, exact and generic pilots, the Arb audit, figures, hashes, and publication artifacts. The main commands are:

```
pytest -q -p no:cacheprovider
python experiments/run_covariance_envelope_pilot.py
python experiments/run_arb_covariance_tradeoff.py
MPLBACKEND=Agg python experiments/make_figures.py
latexmk -pdf -interaction=nonstopmode -halt-on-error main.tex
```

The optimization identities and tradeoff theorem are analytic. The generic model rows are binary64 evidence; the exact cancellation rows have a 256-bit audit. Production covariance derivation, uniform block depth, Stage A1, unconditional Stage A4, a self-adjoint Hilbert–Polya operator, an arithmetic trace formula, and a zeta-zero identity remain open.

References

- [1] Stephen Boyd and Lieven Vandenbergh. *Convex Optimization*. Cambridge University Press, 2004.
- [2] Liang Wang. Block cross-column Krylov gram certificates. Preprint and software repository, 2026. URL https://github.com/maris205/prime_dynamics_theory/tree/main/papers/RH-66-block-cross-column-krylov-gram.